

Two models. One shared ordinance sample.

Lexical TF-IDF + c-TF-IDF vs Legal-BERT + BERTopic

Initial measured results · 200 sections · 2016–2025

Corpuz · Lamsin · Nuril

What this experiment can—and cannot—show

MEASURED NOW

Two complete model paths on identical sections

Two parameter configurations with three seed runs

C_v, diversity, coverage, outliers and stability

Source-linked BERT vectors and yearly visualizations

NOT YET ESTABLISHED

Human-verified corpus or legal topic labels

Held-out generalization or optimal parameters

Expert validation and final thesis conclusions

Population-level changes in municipal policy

Newer methodology governs the implementation

Milestone-1_Corpus-Lamsin-Nuril.pdf

2016–2025; section-level Legal-BERT, UMAP, HDBSCAN, c-TF-IDF; temporal analysis and expert validation.

diagram.png

Explicit Model A lexical path versus Model B embedding path.

Thesis_Corpus, Lamsin, Nuril.pdf

Earlier LDA proposal. Retained as historical context, not the current two-model specification.

Corpus readiness is the largest limitation

7,627 source sections / 1,501 ordinances

7,614 source sections are unverified; 13 verified.

Pilot: 200 sections / 184 ordinances

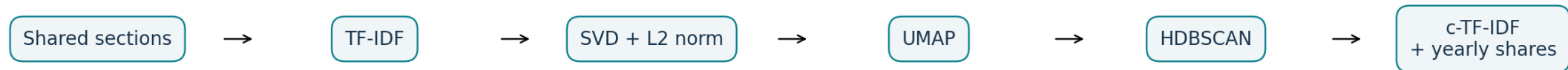
20 seeded random sections per year; same ordered sample for both models.

200 sampled sections remain unverified; 2 exact repeated texts retained.

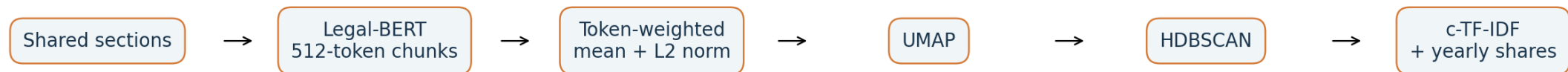
No silent OCR correction, semantic relabeling or invented expert scores.

Shared inputs; representation is the main difference

Model A · Lexical



Model B · Legal-BERT



c-TF-IDF requires existing classes: Model A first clusters TF-IDF/SVD vectors. Both paths then derive topic keywords with c-TF-IDF.

The BERT module: frozen encoder, not fine-tuning

nlpaueb/legal-bert-base-uncased · pinned model revision

Long sections → non-overlapping chunks, up to 512 tokens including special tokens.

Token-weighted mean pooling excludes padding and special tokens.

Aggregate all chunks → L2-normalized section embedding → UMAP → HDBSCAN.

Embedding vectors retain section, ordinance and year associations.

Legal pretraining is not proof of suitability for Philippine municipal law. No learning rate or training epochs apply: encoder weights are frozen.

Recorded parameters—not hypothetical tuning

Parameters actually executed

Trial	UMAP neighbors	UMAP min_dist	HDBSCAN min size	HDBSCAN min samples
initial	10	0.0	5	2
conservative	15	0.1	10	5

Shared: seed 42 | UMAP dimensions 5 | top words 10 | stability runs 3 | no BERT fine-tuning

Pictures are rendered from the executed configuration. Sensitivity trials change several settings together; they do not isolate causal parameter effects.

Initial model comparison

Initial configuration · measured model comparison

Metric	Model A · Lexical	Model B · Legal-BERT
C_v coherence	0.708	0.639
Top-10 diversity	0.839	0.821
Assigned section share	0.955	0.965
Outlier share	0.045	0.035
Seed stability (ARI)	0.804	0.748
Topics (not outliers)	18	14

Higher intrinsic scores do not establish legal validity. No accuracy, F1 or perplexity is claimed.

What the initial numbers suggest—not prove

Coherence: lexical 0.708 vs Legal-BERT 0.639.

Keyword diversity: lexical 0.839 vs Legal-BERT 0.821.

Assigned sections: lexical 95.5% vs Legal-BERT 96.5%.

Seed stability: lexical 0.804 vs Legal-BERT 0.748.

The largest topics contain formal wording such as city, ordinance and attested. Strong intrinsic metrics may reward repeated boilerplate rather than substantive policy themes. More coverage alone does not mean better legal topics.

Treat this as a working baseline for corpus cleanup—not a verdict on either model.

Do results move when clustering settings change?

Measured sensitivity results · same sample

Trial	Model	Topics	C_v	Diversity	Coverage	Seed ARI
initial	lexical	18	0.708	0.839	0.955	0.804
initial	semantic	14	0.639	0.821	0.965	0.748
conservative	lexical	8	0.799	0.800	0.810	0.463
conservative	semantic	3	0.801	0.767	1.000	0.979

No held-out evaluation; no configuration is declared optimal. ARI includes the outlier label as one cluster.

Same input, vocabulary and frozen encoder. Cached BERT embeddings are reused after text-hash validation. Runtime for a cached trial is not an end-to-end speed comparison.

A real Legal-BERT output snippet

```
ENCODER  nlpaueb/legal-bert-base-uncased
REVISION 15b570cbf88259610b082a167dacc190124f60f6
SOURCE   009-16 / section_21 / year 2016
OUTPUT   200 sections x 768 dimensions
VECTOR   [-0.01599  0.0172   0.04295 -0.00549  0.03265  0.00614  0.01445 -0.01639] ...
L2 NORM  1.000000   |   ASSIGNED TOPIC 7
```

Closure and Opening of Roads. (a) A [oca[government unit may, pursuant to an ORDINANCE, permanently or temporarily ctose or open any local' road, a[[ey, park, or square fattig wi.thin its jurisdiction: Provided, however, That ln case of permanent ctosure, such ordinance must be approved by at least two-thirds (2/3)

Representative output: retain source context

Model A · Lexical

T0: attested, secretary, city, correct, enacted

0105-16 / section_4 / 2016

EFFECTMTY- This Ordinance shall take effect immediately upon approval; ENACTED, on the 22nd day of November, by a unanimous vote of all the Members of the Sanggunian present. CERTIFIED CORRECT: "For and in the absence of the Secretary" LMER M. CABALI Acting Secretary the SANGGUNIANG Panlungsod (Superuising Administrative Officerl ATTESTED: DUTERTE P cn Mayor ! t Ord. No. 0105-16 LAr';;'*f A[j't'Hlt SARA Z.

Keywords are machine-generated, not expert-approved policy labels. OCR artifacts and legal boilerplate remain visible.

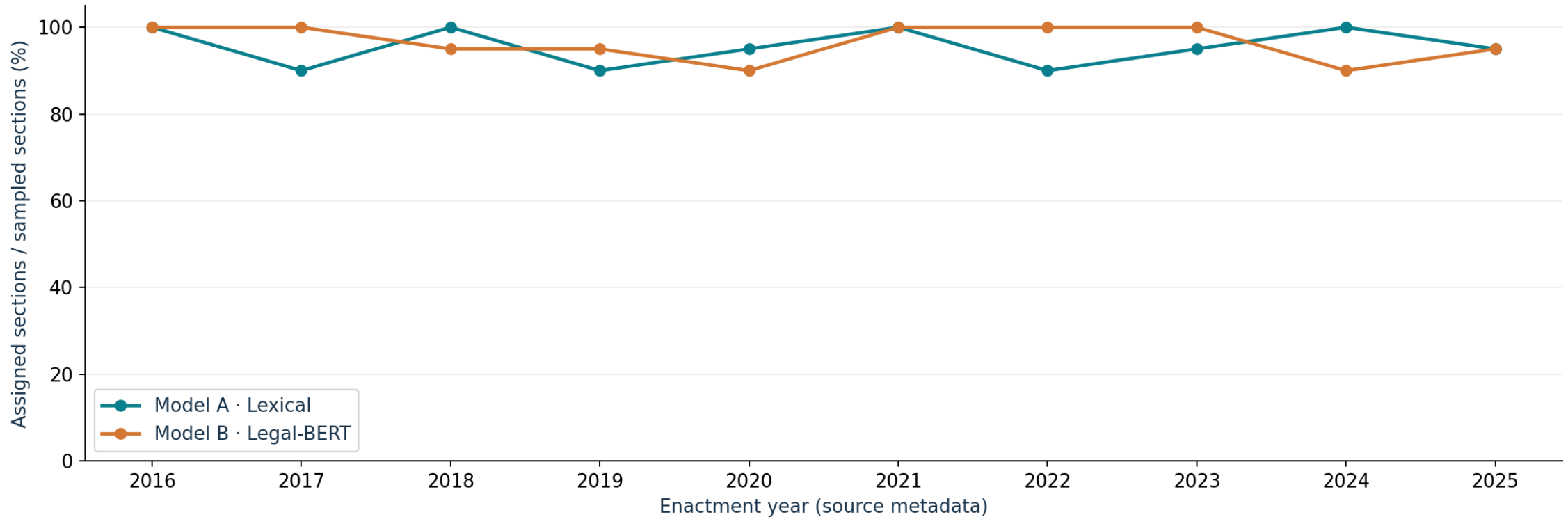
Model B · Legal-BERT

T0: davao, city, ordinance, sign, known

0134-16 / section_1 / 2017

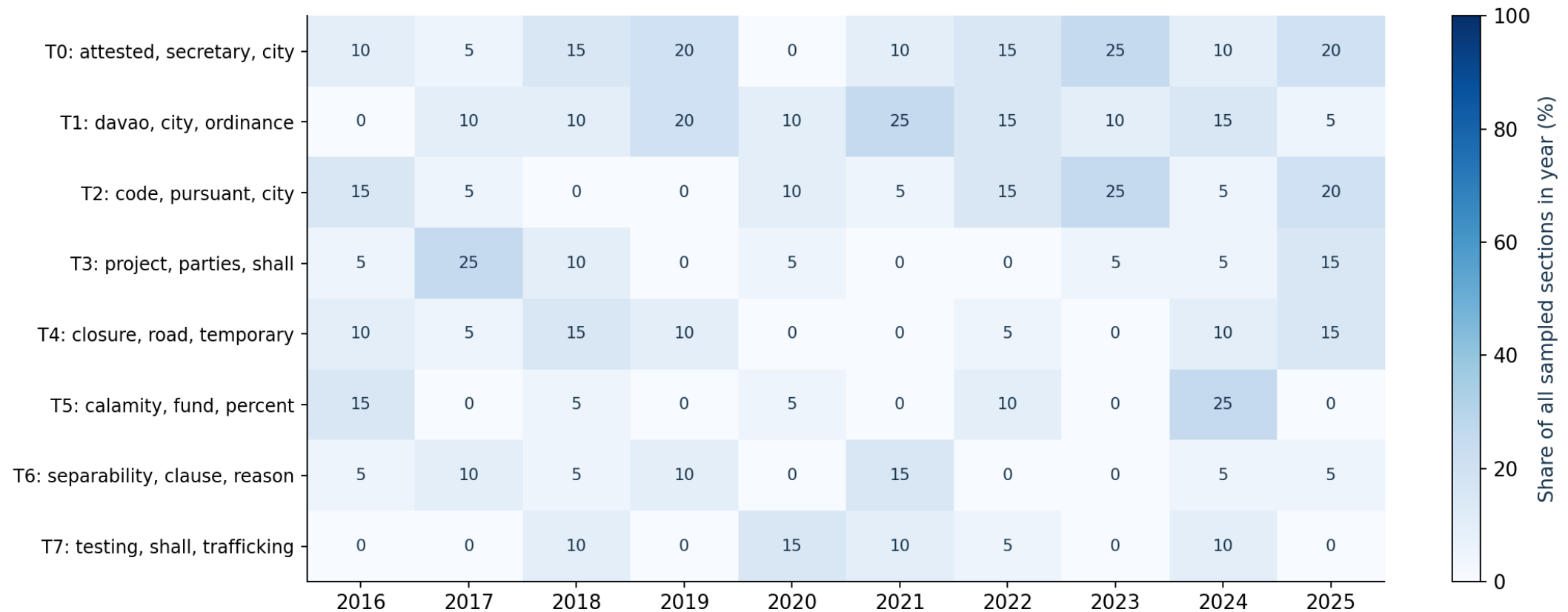
EIE - This Ordinance shall be known as " AN ORDINANCE GRANTING AUTHORIW TO THE CITY MAYOR TO SIGN THE PERFORMANCE COMMITMENT FOR HCIs (REVISION 2) AND PROVIDER DATA RECORD HEALTH CARE INSTITUTION OF TORIL HEATTH CENTER BIRTHING HOME FOR ITS APPLICATION FOR PHILHEATTH ACCREDITATION FOR MATERNITY CARE."

How much of each year's sample gets a topic?



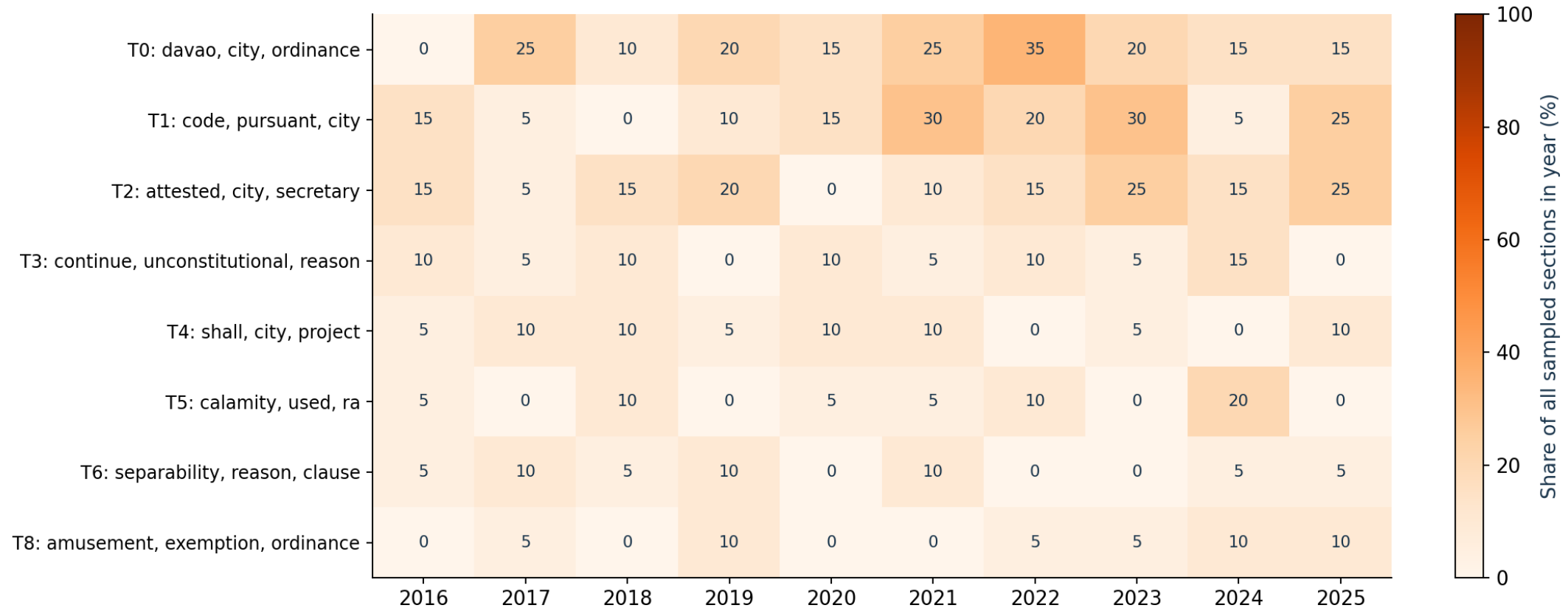
Denominator: all sampled sections in each year, including outliers. Balanced sampling does not estimate ordinance production or full-corpus topic prevalence.

Model A • Lexical • topic shares across years



Top eight global topics only; omitted topics and outliers mean columns need not sum to 100%. These are sample-level patterns, not verified policy trends.

Model B • Legal-BERT • topic shares across years



Top eight global topics only; omitted topics and outliers mean columns need not sum to 100%. These are sample-level patterns, not verified policy trends.

How to read the metrics

C_v: sliding-window coherence of top unigram keywords (Gensim; same input texts).

Diversity: unique topic keywords / all retained top-keyword slots; excludes outliers.

Coverage: assigned sections / all retained sections; outlier share is its complement.

Stability: mean adjusted Rand index vs two additional UMAP seeds; outliers treated as one cluster.

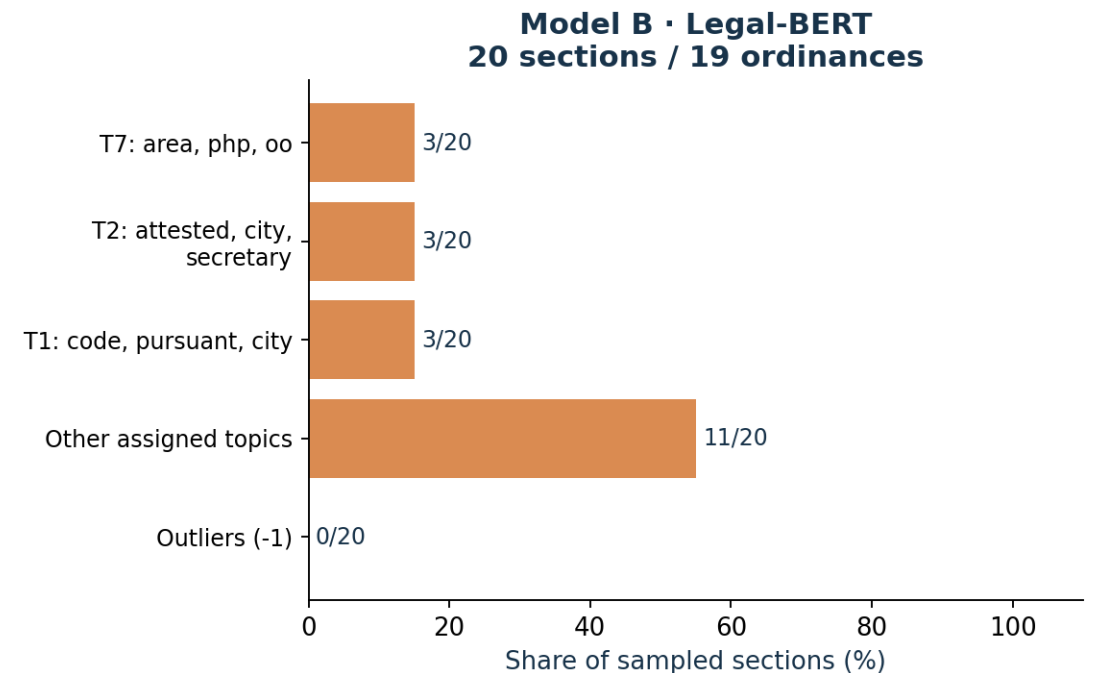
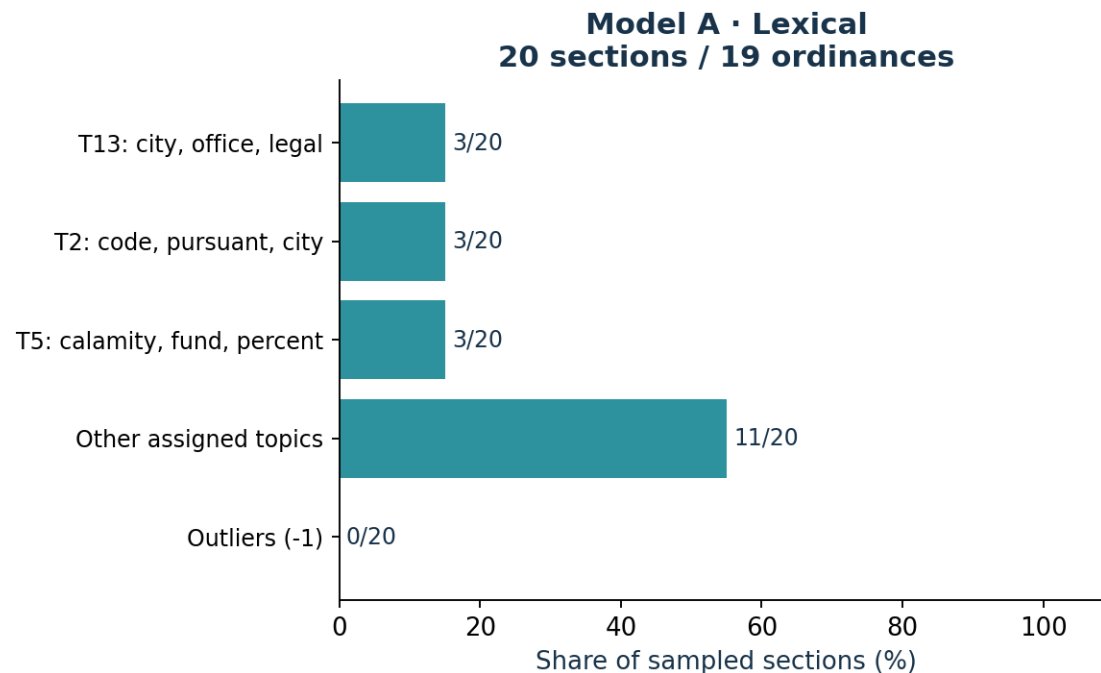
All scores are intrinsic and in-sample. Accuracy/F1 need reference labels. BERTopic has no directly comparable probabilistic perplexity. Expert validation is still pending.

Before these can become thesis findings

- 1 Verify OCR, section boundaries, dates and provenance against the PDFs.
- 2 Review boilerplate, unusually long sections and repeated texts.
- 3 Freeze a verified corpus; rerun both paths on identical inclusion rules.
- 4 Use ordinance-grouped development / held-out evaluation before final selection.
- 5 Expert panel: legal coherence, relevance, label appropriateness, temporal interpretation, overall quality (1–5 / N/A).
- 6 Review any expert dimension below 3.00; regenerate the presentation from new evidence.

2016 • both models, side by side

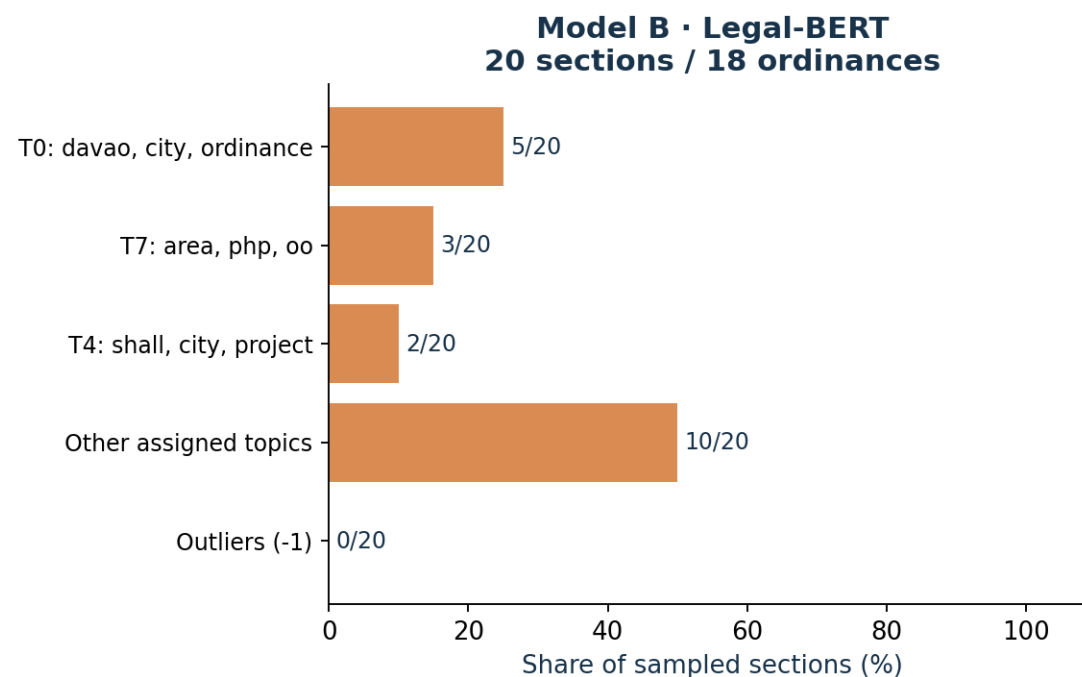
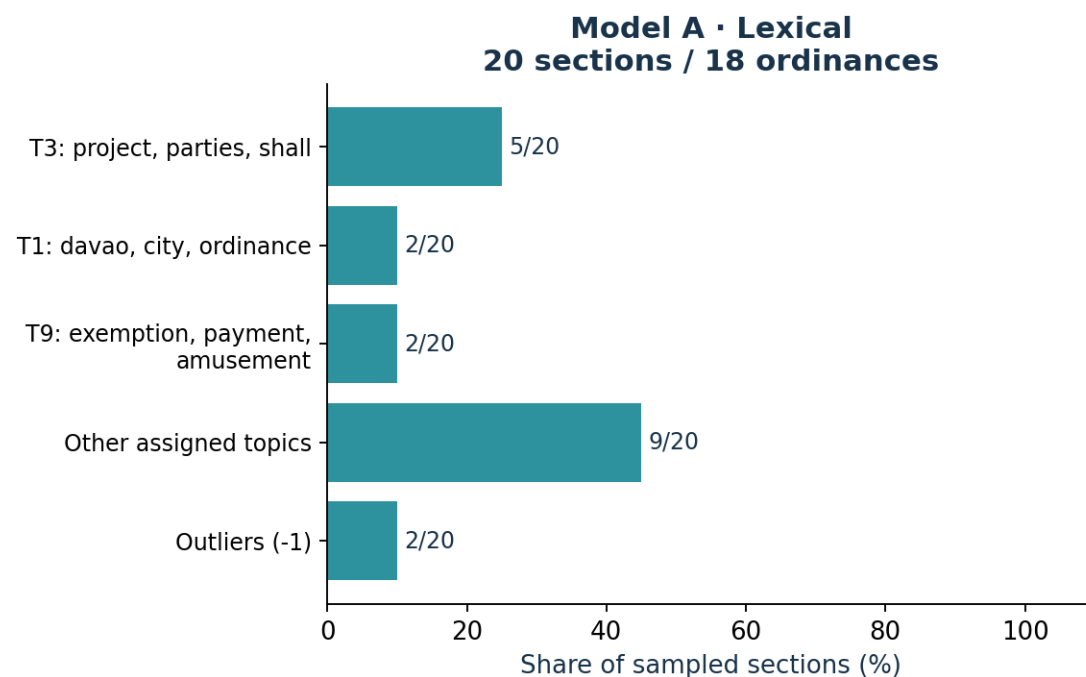
2016 • exploratory within-year comparison



Topic numbers are local to each model: A/T0 is not B/T0. Bars show the top three topics, other assigned topics and outliers within this year's sample.

2017 • both models, side by side

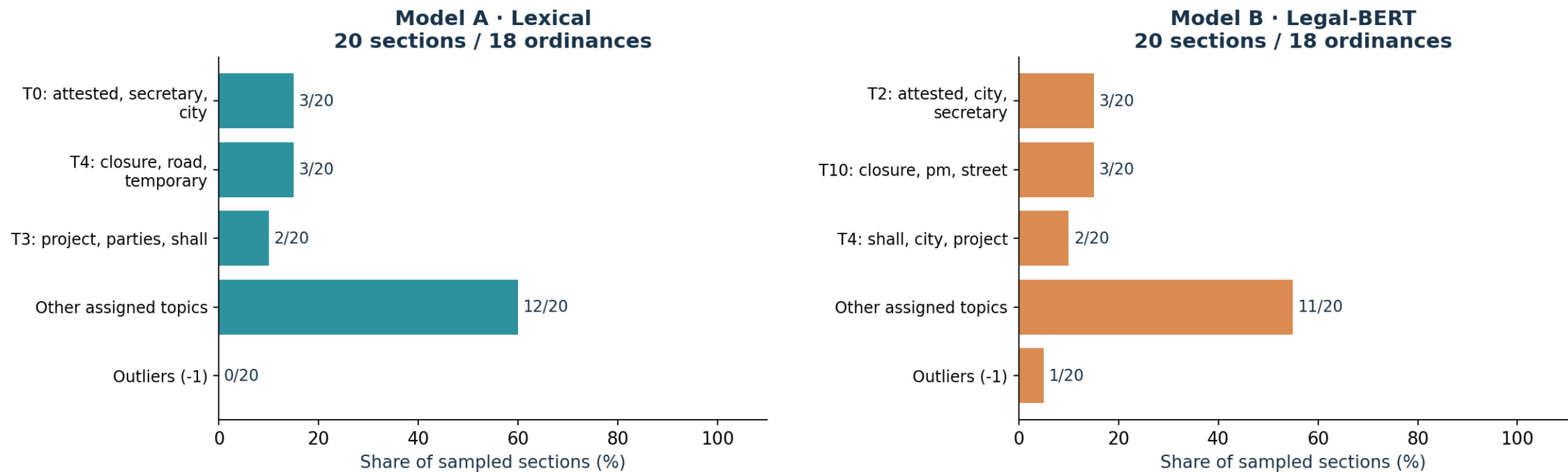
2017 • exploratory within-year comparison



Topic numbers are local to each model: A/T0 is not B/T0. Bars show the top three topics, other assigned topics and outliers within this year's sample.

2018 · both models, side by side

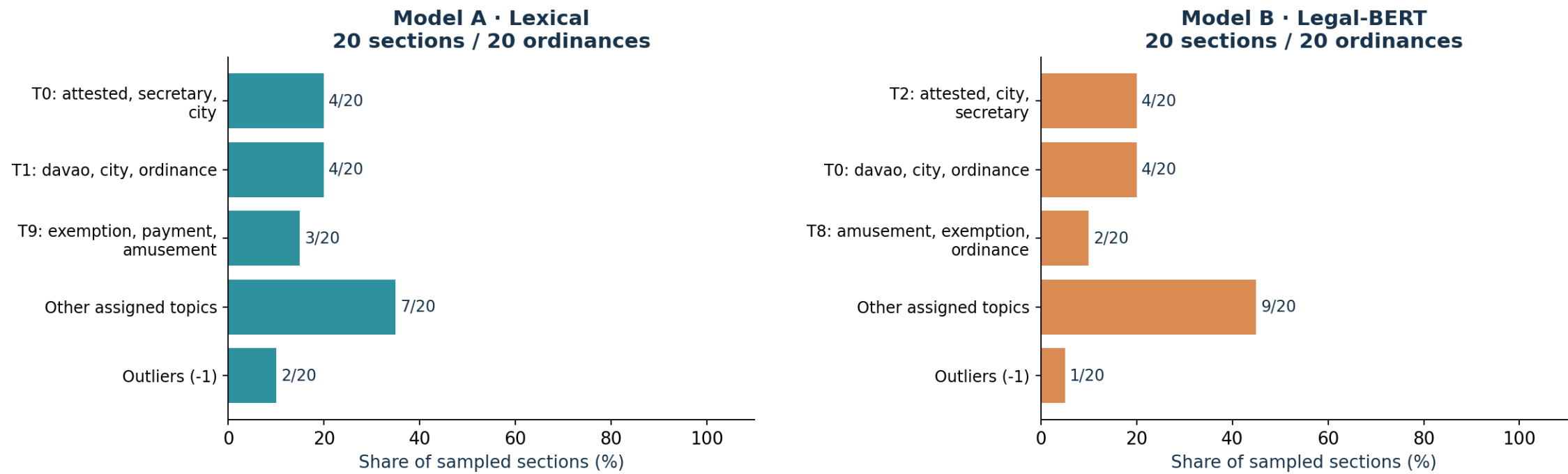
2018 · exploratory within-year comparison



Topic numbers are local to each model: A/T0 is not B/T0. Bars show the top three topics, other assigned topics and outliers within this year's sample.

2019 · both models, side by side

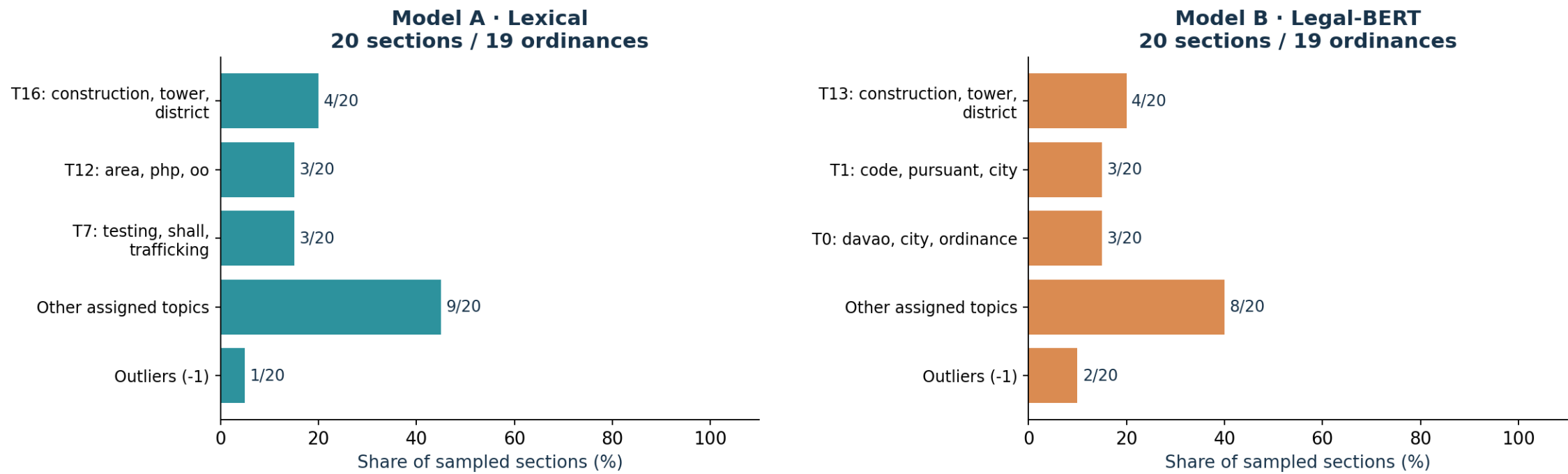
2019 · exploratory within-year comparison



Topic numbers are local to each model: A/T0 is not B/T0. Bars show the top three topics, other assigned topics and outliers within this year's sample.

2020 • both models, side by side

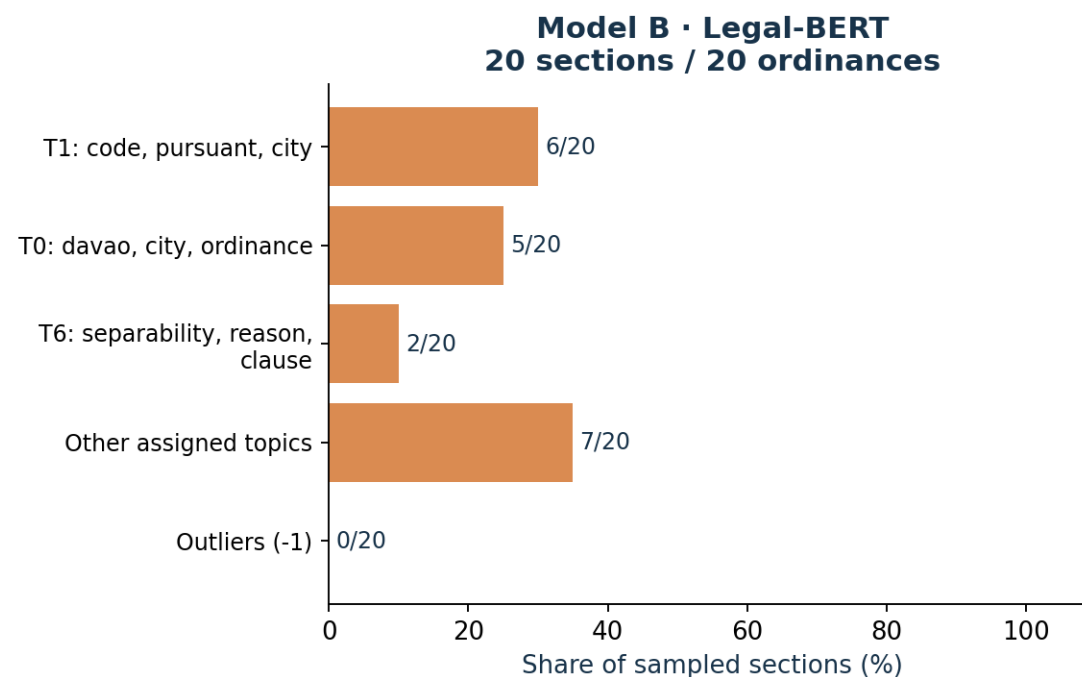
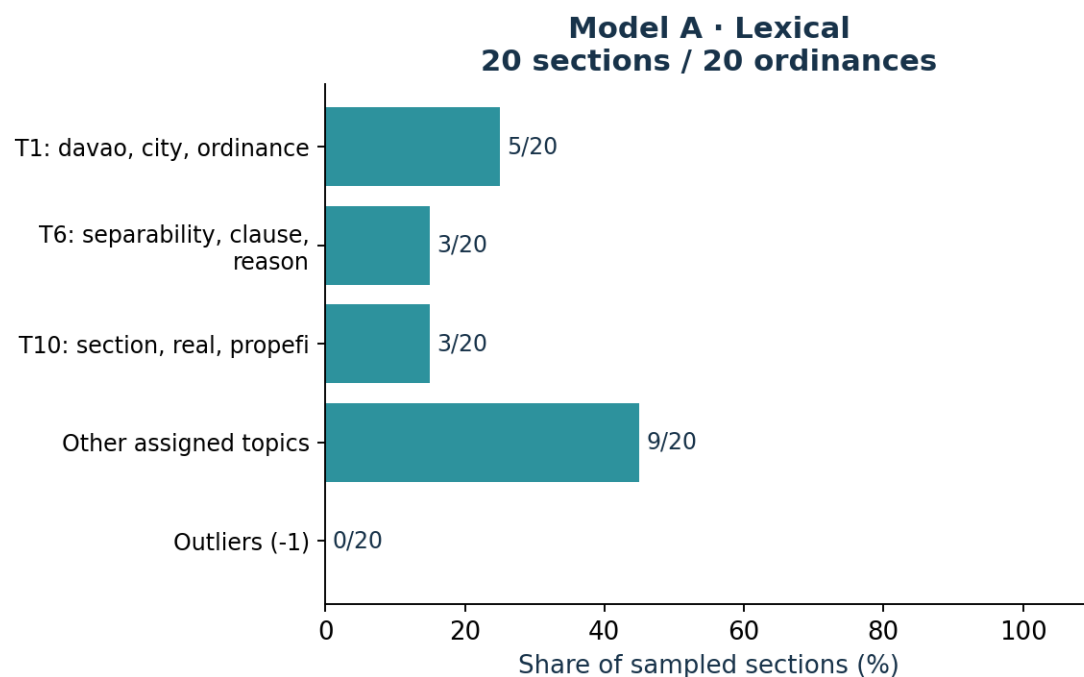
2020 • exploratory within-year comparison



Topic numbers are local to each model: A/T0 is not B/T0. Bars show the top three topics, other assigned topics and outliers within this year's sample.

2021 · both models, side by side

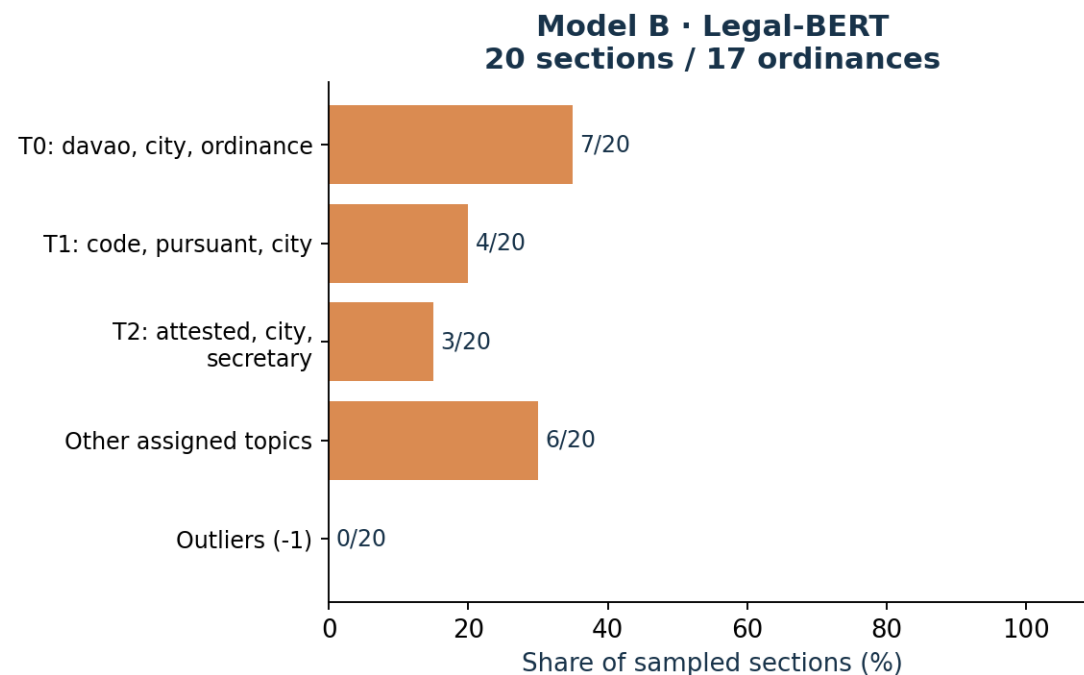
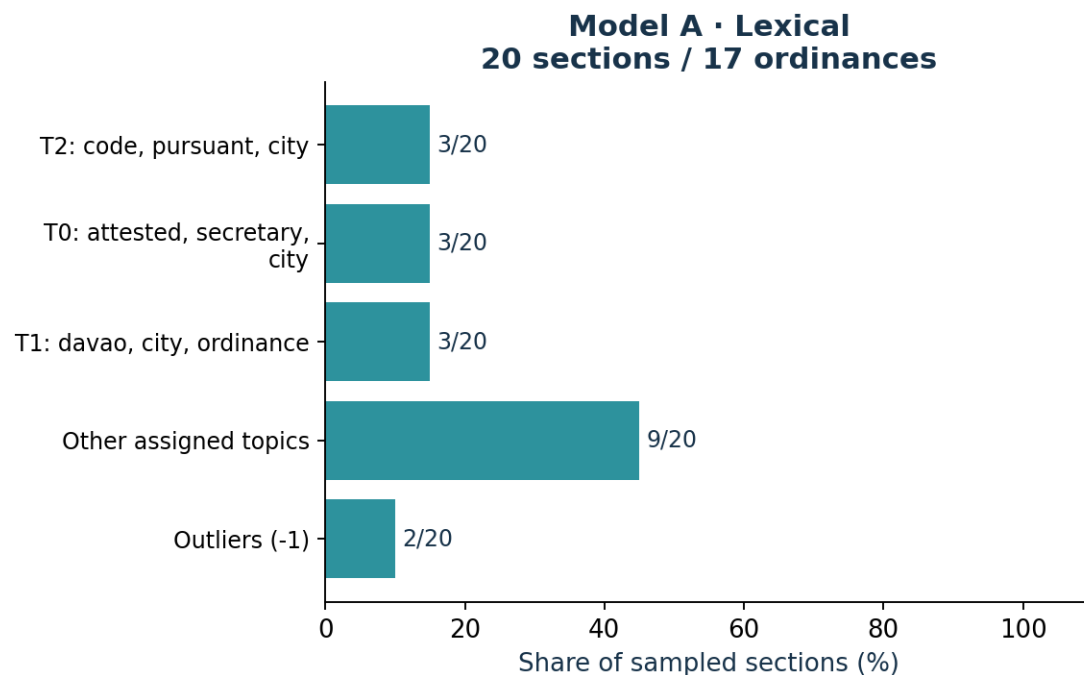
2021 · exploratory within-year comparison



Topic numbers are local to each model: A/T0 is not B/T0. Bars show the top three topics, other assigned topics and outliers within this year's sample.

2022 • both models, side by side

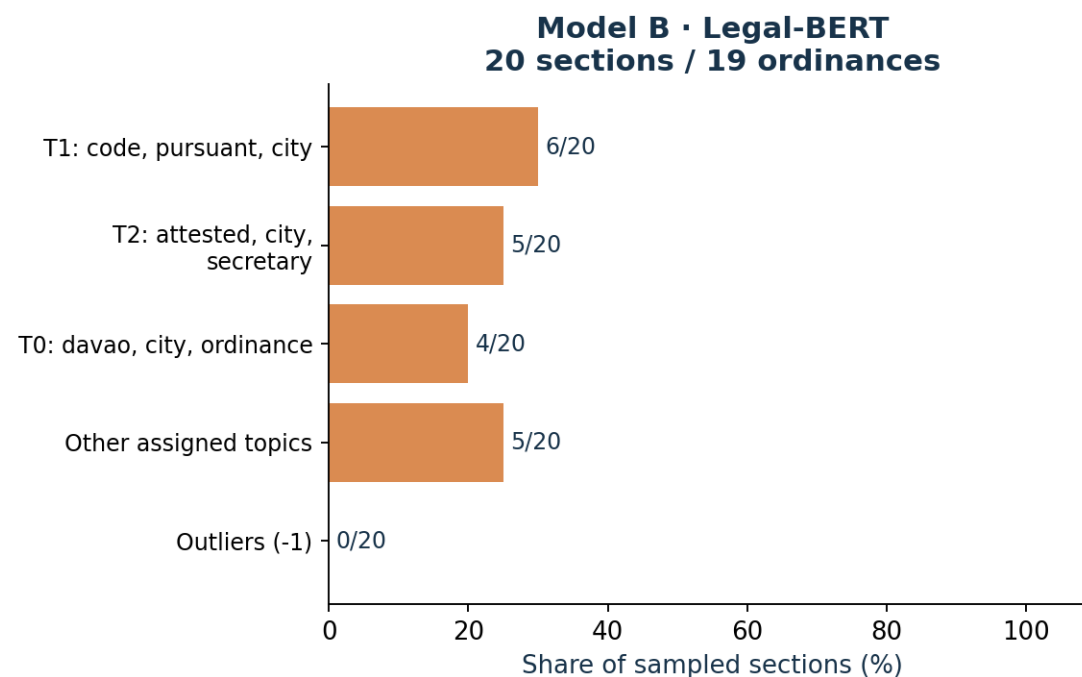
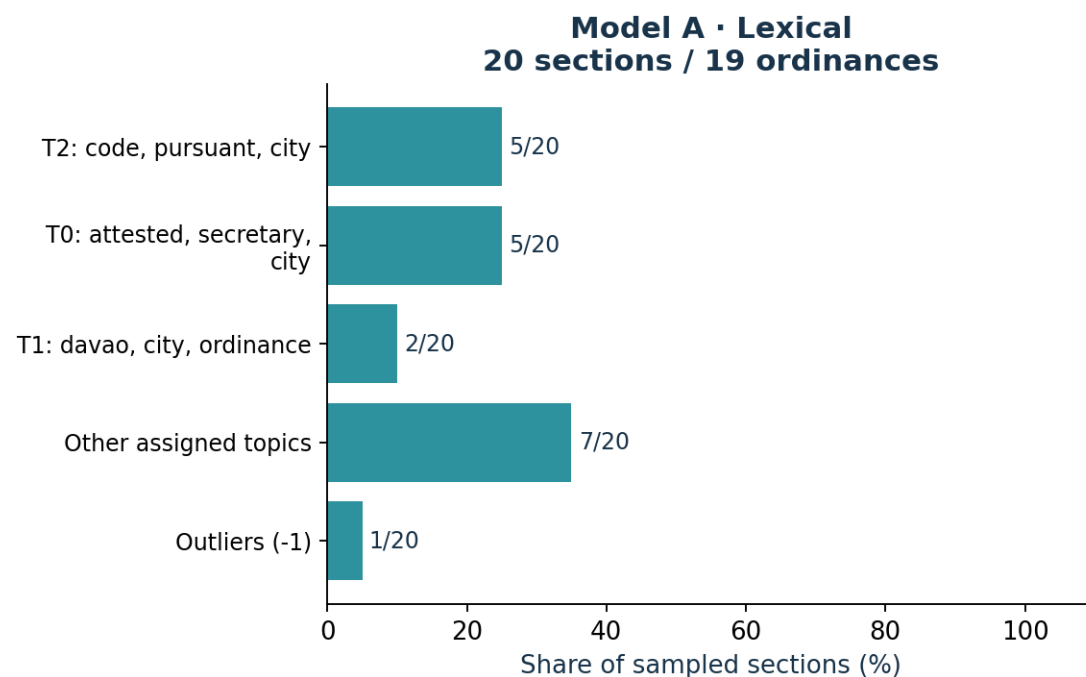
2022 • exploratory within-year comparison



Topic numbers are local to each model: A/T0 is not B/T0. Bars show the top three topics, other assigned topics and outliers within this year's sample.

2023 · both models, side by side

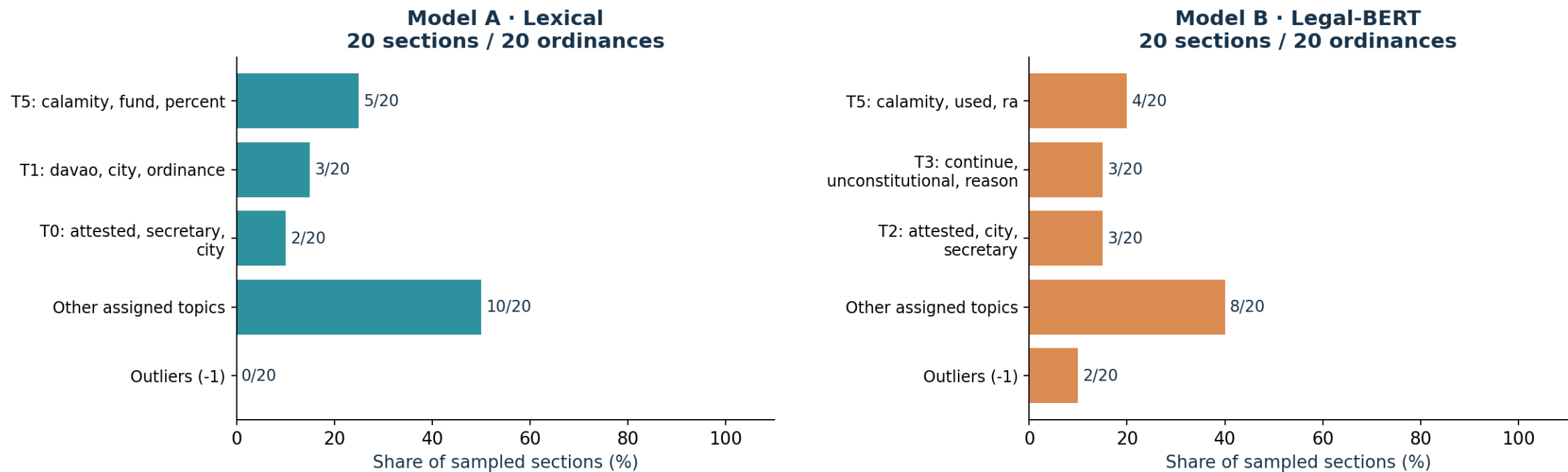
2023 · exploratory within-year comparison



Topic numbers are local to each model: A/T0 is not B/T0. Bars show the top three topics, other assigned topics and outliers within this year's sample.

2024 · both models, side by side

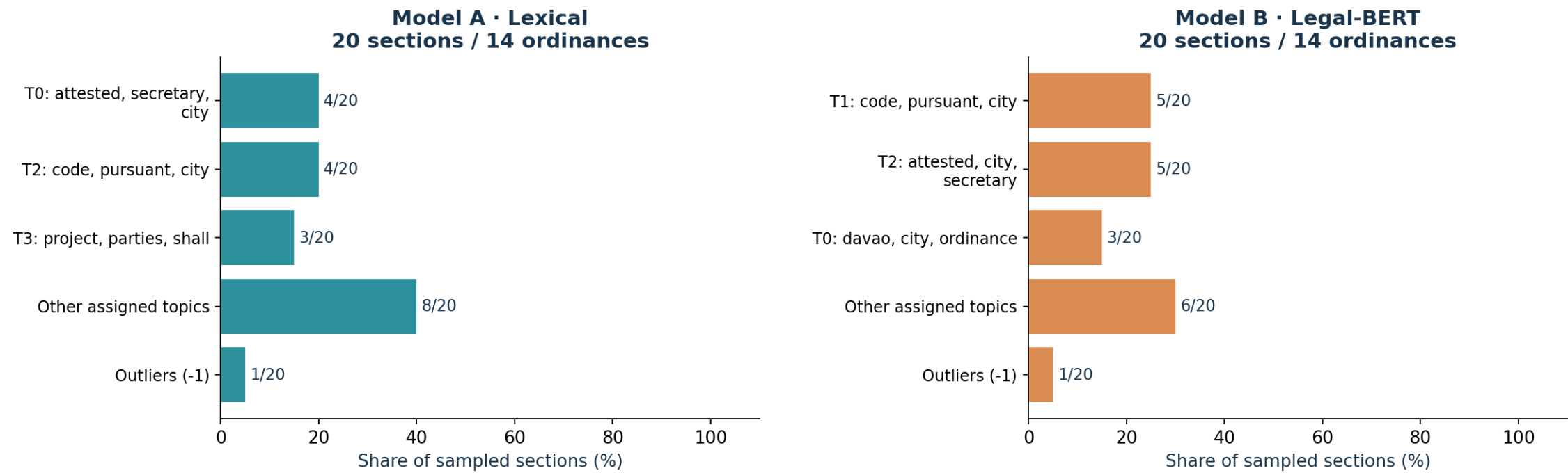
2024 · exploratory within-year comparison



Topic numbers are local to each model: A/T0 is not B/T0. Bars show the top three topics, other assigned topics and outliers within this year's sample.

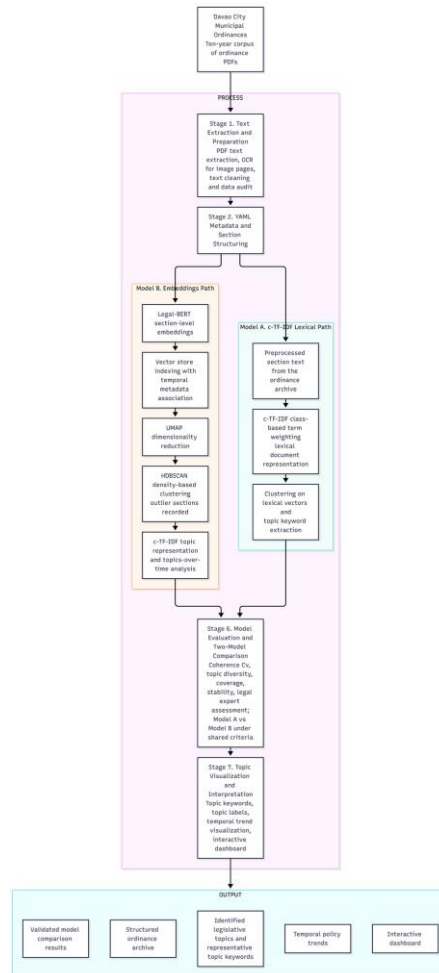
2025 · both models, side by side

2025 · exploratory within-year comparison



Topic numbers are local to each model: A/T0 is not B/T0. Bars show the top three topics, other assigned topics and outliers within this year's sample.

Original supplied process diagram



Source: Downloads/diagram.png

Implementation clarification:

c-TF-IDF needs class assignments. The lexical path therefore uses TF-IDF/SVD before clustering, then c-TF-IDF for topic representation.

The current deliverable is an exploratory modeling and reporting pipeline—not a completed expert-validated system.